Google Cloud

Google Cloud Engineering Certificate

2025

Proprietary & Confidential

Google Cloud

Why Cloud Engineering?

Cloud engineering provides a scalable way to design and build a cost-effective digital infrastructure. By utilizing cloud services, you can drive innovation and create reliable solutions to stay ahead in an environment that is rapidly evolving. Whether you're looking to launch your career or push your organization forward, cloud engineering positions you at the forefront of technological transformation.

Google Cloud Engineering Certificate

This certificate program is designed for learners looking to build expertise in cloud infrastructure and prepare for careers in a hybrid cloud world. A Cloud Engineer plans, configures, sets up, and deploys cloud solutions. A college degree is not a prerequisite, but learners should be familiar with operating systems, file systems, the IP networking stack, how web browsers and web servers communicate, databases, programming, working with Linux and editing text files from the command line. The content provides learners with intermediate to advanced skills in cloud-based architecture.

There are 6 courses in the Google Cloud Engineering Certificate. In each course, learners will complete hands-on labs that allow them to practice key skills and produce work examples to show potential employers. The first 6 courses progressively build learners' working knowledge and the final 4 lab-based courses provide the learner with hands-on practice and skill badge credentials.

6 courses

38 modules

207 videos

23 labs

20 documents

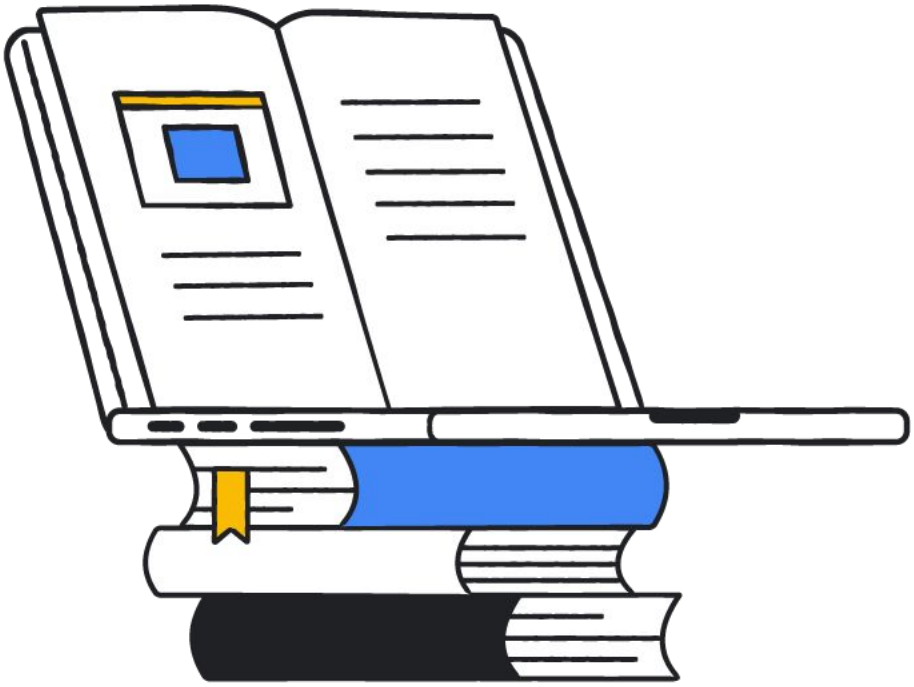
31 quizzes

4 skill badges

~ 70 hours to complete

Google Cloud

Courses



Certificate

1. Preparing for Your Associate Cloud Engineer Journey

Course Overview

This course helps you structure your preparation for the Associate Cloud Engineer exam. You will learn about the Google Cloud domains covered by the exam and how to create a study plan to improve your domain knowledge.

Prerequisites (optional):

Familiarity with Google Cloud, the Linux command line, web servers, and text editors

By the end of this course, learners will:

- Describe the five domains covered on the Associate Cloud Engineer Certification exam.
- Identify gaps in your knowledge and skills for each domain and create a study plan.

Course Outline

Module 1: Introduction
Module 2: Setting Up a Cloud Solution Environment
Module 3: Planning and Configuring a Cloud Solution
Module 4: Deploying and Implementing a Cloud Solution
Module 5: Ensuring Successful Operation of a Cloud Solution
Module 6: Configuring Access and Security
Module 7: Your next steps

~ 8 hours to complete

Certificate

2. Google Cloud Fundamentals: Core Infrastructure

Course Overview

This course introduces important concepts and terminology for working with Google Cloud. Through videos and hands-on labs, this course presents and compares many of Google Cloud's computing and storage services, along with important resource and policy management tools.

By the end of this course, learners will:

- Identify the purpose and value of Google Cloud products and services
- Define how infrastructure is organized and controlled in Google Cloud
- Explain how to create a basic infrastructure in Google Cloud
- Select and use Google Cloud storage options

Course Outline

Module 0: Course Introduction
Module 1: Introducing Google Cloud
Module 2: Resources and Access in the Cloud
Module 3: Virtual Machines and Networks in the Cloud
Module 4: Storage in the Cloud
Module 5: Containers in the Cloud
Module 6: Applications in the Cloud
Module 7: Course Summary

~ 6 hours to complete

Certificate

3. Essential Google Cloud Infrastructure: Foundation

Course Overview

This accelerated course introduces participants to the comprehensive and flexible infrastructure and platform services provided by Google Cloud with a focus on Compute Engine. You will explore and deploy solution elements, including infrastructure components such as networks, virtual machines and applications services. You will learn how to use the Google Cloud through the console and Cloud Shell. You'll also learn about the role of a cloud architect, approaches to infrastructure design, and virtual networking configuration with Virtual Private Cloud (VPC), Projects, Networks, Subnetworks, IP addresses, Routes, and Firewall rules.

Prerequisites:

- Completed Google Cloud Fundamentals: Core Infrastructure or have equivalent experience.
- Have basic proficiency with command-line tools and Linux operating system environments.
- Have systems operations experience, including deploying and managing applications, either on-premises or in a public cloud environment.

By the end of this course, learners will:

- Interact with the Google Cloud console and Cloud Shell
- Deploy solutions using Google Cloud Marketplace
- Implement VPC networks and firewall rules
- Create and customize VM instances using Compute Engine

Course Outline

Module 0: Introduction
Module 1: Interacting with Google Cloud
Module 2: Virtual Networks
Module 3: Virtual Machines
Module 4: Course Resources

~ 9.5 hours to complete

Certificate

4. Essential Google Cloud Infrastructure: Core Services

Course Overview

This accelerated course introduces participants to the comprehensive and flexible infrastructure and platform services provided by Google Cloud with a focus on Compute Engine. Through a combination of video lectures, demos, and hands-on labs, participants explore and deploy solution elements, including infrastructure components such as networks, systems and applications services. This course also covers deploying practical solutions including customer-supplied encryption keys, security and access management, quotas and billing, and resource monitoring.

Prerequisites:

- Have completed Google Cloud Fundamentals: Core Infrastructure or have equivalent experience.
- Have basic proficiency with command-line tools and Linux operating system environments.
- Have systems operations experience, including deploying and managing applications, either on-premises or in a public cloud environment.

By the end of this course, learners will:

- Interact with the Google Cloud console and Cloud Shell
- Deploy solutions using Google Cloud Marketplace
- Implement VPC networks and firewall rules
- Create and customize VM instances using Compute Engine

Course Outline

Module 0: Course Introduction
Module 1: Identity and Access Management (IAM)
Module 2: Storage and Database Services
Module 3: Resource Management
Module 4: Resource Monitoring
Module 5: Course Resources

~ 9 hours to complete

Certificate

5. Essential Google Cloud Infrastructure: Scaling and Automation

Course Overview

This accelerated course introduces participants to the comprehensive and flexible infrastructure and platform services provided by Google Cloud. Through a combination of video lectures, demos, and hands-on labs, participants explore and deploy solution elements, including securely interconnecting networks, load balancing, autoscaling, infrastructure automation and managed services.

Prerequisites:

- Have completed Google Cloud Fundamentals: Core Infrastructure or have equivalent experience.
- Have basic proficiency with command-line tools and Linux operating system environments.
- Have systems operations experience, including deploying and managing applications, either on-premises or in a public cloud environment.

By the end of this course, learners will:

- Administer Identity and Access Management for resources.
- Implement data storage services in Google Cloud
- Manage and examine billing of Google Cloud resources.
- Monitor resources using Google Cloud's operations suite

Course Outline

Module 0: Introduction
Module 1: Interconnecting Networks
Module 2: Load Balancing and Autoscaling
Module 3: Infrastructure Automation
Module 4: Managed Services
Module 5: Course Resources

~ 10 hours to complete

Certificate

6. Getting Started with Google Kubernetes Engine

Course Overview

The goal of this course is to introduce the basics of Google Kubernetes Engine, or GKE, as it's commonly referred to, and how to get applications containerized and running in Google Cloud. The course starts with a basic introduction to Google Cloud, and is then followed by an overview of containers and Kubernetes, Kubernetes architecture, and Kubernetes operations.

Prerequisites:

- Learners should have a basic proficiency with command-line tools, Linux operating system environments, and web server technologies like Nginx.
- Experience with systems operations, including deploying and managing applications, either on-premises or in a public cloud environment is recommended.

By the end of this course, learners will:

- Discuss the differences among Google Cloud compute platforms.
- Discuss the components and architecture of Kubernetes.
- Identify how Google manages Kubernetes orchestration.
- Create and manage Google Kubernetes Engine clusters by using the Google Cloud console and gcloud/kubectl commands.

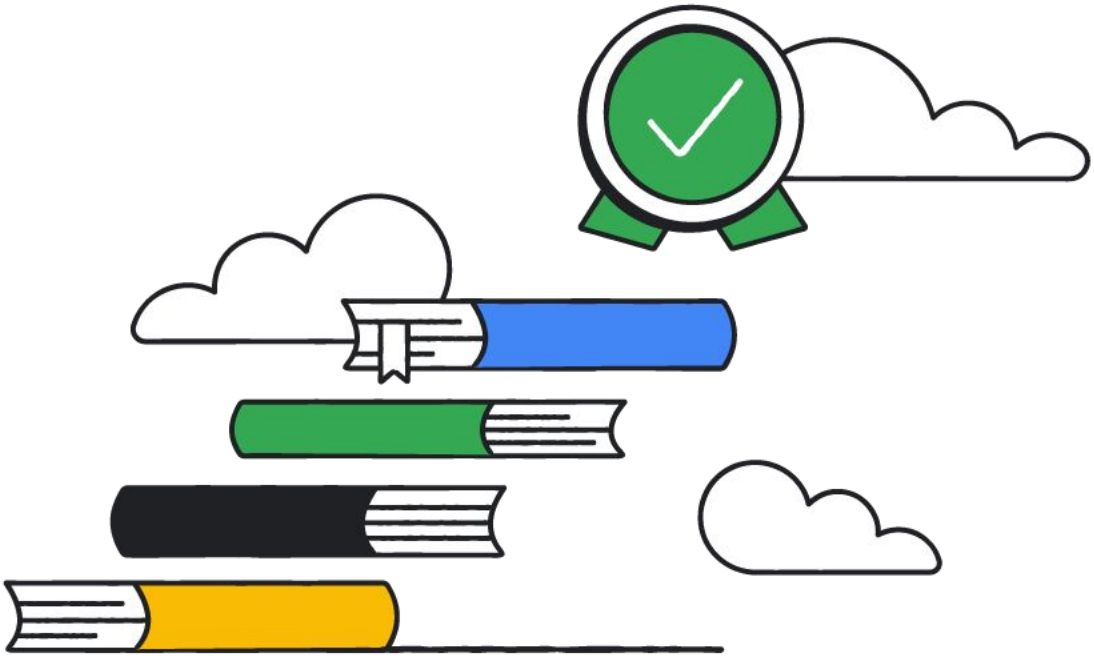
Course Outline

Module 0: Course Introduction
Module 1: Introduction to Google Cloud
Module 2: Introduction to Containers and Kubernetes
Module 3: Kubernetes Architecture
Module 4: Kubernetes Operations
Module 5: Course Summary

~ 7 hours to complete

Google Cloud

Google Cloud Skill Badges



Skill Badge

1. Implement Load Balancing on Compute Engine

Skills Badge Overview

Complete the introductory Implement Load Balancing on Compute Engine skill badge* to demonstrate skills in the following: writing gcloud commands and using Cloud Shell, creating and deploying virtual machines in Compute Engine, and configuring network and HTTP load balancers.

Course Outline

Lab 1

- Creating a Virtual Machine: In this hands-on lab, you create VM instances of various machine types using the Google Cloud console and the gcloud command line in Cloud Shell. You also learn how to connect an NGINX web server to your VM.

  OR

- Compute Engine: Qwik Start - Windows: In this hands-on lab, you learn how to launch a Windows Server instance in Compute Engine and use Remote Desktop Protocol (RDP) to connect to it.

Lab 2

- Getting Started with Cloud Shell and gcloud: In this hands-on lab, you learn how to connect to computing resources hosted on Google Cloud via Cloud Shell with the gcloud tool.

Lab 3

- Set Up Network and HTTP Load Balancers: In this hands-on lab you learn the differences between a network load balancer and an HTTP load balancer, and how to set them up for your applications running on Compute Engine virtual machines (VMs).

Challenge Lab - Implement Load Balancing on Compute Engine: In a challenge lab you're given a scenario and a set of tasks. You will use the skills learned from the labs in the course to figure out how to complete the tasks on your own!

*A skill badge is an exclusive digital badge issued by Google Cloud in recognition of your proficiency with Google Cloud products and services and tests your ability to apply your knowledge in an interactive hands-on environment. Complete this skill badge, and the final assessment challenge lab, to receive a skill badge that you can share with your network.

~ 3.5 hours to complete

Skill Badge

2. Set Up an App Dev Environment on Google Cloud

Skills Badge Overview

Earn a skill badge* by completing the Set Up an App Dev Environment on Google Cloud course, where you learn how to build and connect storage-centric cloud infrastructure using the basic capabilities of the of the following technologies: Cloud Storage, Identity and Access Management, Cloud Functions, and Pub/Sub.

Course Outline

Lab 1

- Cloud Storage: Qwik Start - Cloud Console: In this hands-on lab you will learn how to use the Cloud console to create a storage bucket, upload objects to the bucket, create folders and subfolders in the bucket, and make objects in a storage bucket publicly accessible.

  OR

- Cloud Storage: Qwik Start - CLI/SDK: In this hands-on lab you will learn how to create a storage bucket, upload objects to it, create folders and subfolders in it, and make objects publicly accessible using the Google Cloud command line.

Lab 2

- Cloud IAM: Qwik Start: In this lab you'll sign in with 2 different sets of credentials to experience how granting and revoking permissions works from Google Cloud Project Owner and Viewer roles.

Lab 3

- Cloud Monitoring: Qwik Start: In this lab you'll install monitoring and logging agents to collect information from your instance, which could include metrics and logs from 3rd party apps.

Lab 4

- Cloud Functions: Qwik Start - Console: This hands-on lab shows you how to create, deploy, and test a cloud function using the Google Cloud console.

  OR

- Cloud Functions: Qwik Start - Command Line: This hands-on lab shows you how to create, deploy, and test a cloud function using the Google Cloud Shell command line.

Skill Badge

2. Set Up an App Dev Environment on Google Cloud

Lab 5

- Pub/Sub: Qwik Start - Console: In this hands-on lab you will learn how to set up a topic to hold data, subscribe to a topic to access the data and publish and then consume messages with a pull subscriber.

  OR

- Cloud Storage: Qwik Start - CLI/SDK: In this hands-on lab you will learn how to create a storage bucket, upload objects to it, create folders and subfolders in it, and make objects publicly accessible using the Google Cloud command line.

  OR

- Pub/Sub: Qwik Start - Python: In this lab, you will learn how to get started publishing messages with Pub/Sub using the Python client library.

Challenge Lab - Set Up an App Dev Environment on Google Cloud: In a challenge lab you're given a scenario and a set of tasks. You will use the skills learned from the labs in the course to figure out how to complete the tasks on your own!

*A skill badge is an exclusive digital badge issued by Google Cloud in recognition of your proficiency with Google Cloud products and services and tests your ability to apply your knowledge in an interactive hands-on environment. Complete this skill badge, and the final assessment challenge lab, to receive a skill badge that you can share with your network.

~ 4 hours to complete

3. Develop your Google Cloud Network

Skills Badge Overview

Earn a skill badge by completing the Develop your Google Cloud Network course, where you learn multiple ways to deploy and monitor applications including how to: explore IAM roles and add/remove project access, create VPC networks, deploy and monitor Compute Engine VMs, write SQL queries, deploy and monitor VMs in Compute Engine, and deploy applications using Kubernetes with multiple deployment approaches.

Course Outline

Lab 1

- Cloud IAM: Qwik Start: In this lab you'll sign in with 2 different sets of credentials to experience how granting and revoking permissions works from Google Cloud Project Owner and Viewer roles.

Lab 2

- Introduction to SQL for BigQuery and Cloud SQL: This lab is divided into two parts: in the first half, you will learn fundamental SQL querying keywords. In the second half, you will learn how to export subsets of the dataset into CSV files, upload to Cloud SQL and then use Cloud SQL to create and manage databases and tables.

Lab 3

- Multiple VPC Networks: In this lab you create several VPC networks and VM instances, then test connectivity across networks.

Lab 4

- Cloud Monitoring: Qwik Start: In this lab you'll install monitoring and logging agents to collect information from your instance, which could include metrics and logs from 3rd party apps.

Skill Badge

3. Develop your Google Cloud Network

Lab 5

- Managing Deployments Using Kubernetes Engine: This lab teaches you how to scale and manage containers so you can accomplish these common scenarios where multiple heterogeneous deployments are being used.

Challenge Lab - Develop your Google Cloud Network: In a challenge lab you're given a scenario and a set of tasks. You will use the skills learned from the labs in the course to figure out how to complete the tasks on your own!

*A skill badge is an exclusive digital badge issued by Google Cloud in recognition of your proficiency with Google Cloud products and services and tests your ability to apply your knowledge in an interactive hands-on environment. Complete this skill badge, and the final assessment challenge lab, to receive a skill badge that you can share with your network.

~ 6 hours to complete

Skill Badge

4. Build Infrastructure with Terraform on Google Cloud

Skills Badge Overview

Complete the Build Infrastructure with Terraform on Google Cloud skill badge to demonstrate skills in the following: Infrastructure as Code (IaC) principles using Terraform, provisioning and managing Google Cloud resources with Terraform configurations, effective state management (local and remote), and modularizing Terraform code for reusability and organization.

Course Outline

Lab 1

- Terraform Fundamentals: In this lab, you will learn how to get started with Terraform in Google Cloud, install Terraform from installation binaries and create a VM instance infrastructure using Terraform.

Lab 2

- Infrastructure as Code with Terraform: In this lab, you will learn how to build, change, and destroy infrastructure with Terraform, create Resource Dependencies with Terraform, and provision infrastructure with Terraform

Lab 3

- Interact with Terraform Modules: In this lab, you will learn how modules can address these problems, the structure of a Terraform module, and best practices when using and creating modules.

Lab 4

- Managing Terraform State: In this lab, you will learn how to create a local and Cloud Storage backend, refresh your Terraform state, import a Terraform configuration and manage the imported configuration with Terraform.

Challenge Lab - Build Infrastructure with Terraform on Google Cloud: In a challenge lab you're given a scenario and a set of tasks. You will use the skills learned from the labs in the course to figure out how to complete the tasks on your own!

*A skill badge is an exclusive digital badge issued by Google Cloud in recognition of your proficiency with Google Cloud products and services and tests your ability to apply your knowledge in an interactive hands-on environment. Complete this skill badge, and the final assessment challenge lab, to receive a skill badge that you can share with your network.

~ 5 hours to complete

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